Topic 4 – Digital technologies

4.1 CAD, CAM and rapid prototyping

Spec: CAD – 2D and 3D design to create and modify designs and create simulations, 3D modelling for creating 'virtual' products; CAM and rapid prototyping – CNC lathes, CNC routers, CNC milling machine, CNC laser, CNC vinyl cutters, rapid prototyping; Set-up, safe and accurate operation, advantages and disadvantages

Key facts

• **Advantages of CAD over hand drawing?** Accurate and repeatable, easy to edit/copy/mirror/scale, photorealistic renders, virtual testing (FEA, simulation), files shared instantly worldwide, link directly to CAM, libraries of standard parts.
• **Disadvantages of CAD?** Expensive software/hardware and training, regular updates, file compatibility issues, risk of data loss/viruses, slower for quick ideas, can de-skill.
• **What is a CNC router used for?** Cutting 2D/2.5D shapes, profiles, pockets and engravings in wood, MDF, plastics, foam, soft metals.
• **CNC laser cutter – uses and materials?** Cutting and engraving acrylic, plywood, MDF, card, fabric, leather; not PVC (toxic chlorine gas) or polycarbonate.
• **CNC vinyl cutter – use?** A drag knife cuts self-adhesive vinyl (cuts through vinyl but not the backing) for signs, decals, helmet graphics, stencils.
• **Checks before vinyl cutting?** Correct colour/type of vinyl, free from creases/damage, enough width/length loaded, blade depth and pressure set (2019 Q1a).
• **Name three rapid prototyping methods.** FDM (fused deposition modelling – melted filament), SLA (stereolithography – UV laser cures liquid resin), SLS (selective laser sintering – laser fuses powder).
• **Benefits of rapid prototyping?** Fast models for testing fit/form/ergonomics, complex shapes in one operation, cheaper iterations, reduces time to market.
• **Drawbacks of rapid prototyping?** Slow for large parts, limited materials and strength, layer lines need finishing, not usually suitable for mass production.
• **Set-up and safe operation of a laser cutter?** Check material is safe, focus the lens, set power/speed, secure sheet, extraction on, lid closed, never leave unattended (fire), know the emergency stop.
• **CNC lathe vs CNC mill?** Lathe: workpiece rotates, tool moves (cylindrical parts). Mill: cutter rotates, workpiece/table moves in X/Y/Z (flat faces, slots, pockets, 3D forms).

Advantages	Disadvantages
Accurate, repeatable, easily edited (copy, mirror, array, scale)	High cost of software, hardware and training
Photorealistic virtual products reduce need for physical models	Software updates, licences and technical support
Virtual testing saves time and money (FEA, simulation)	File compatibility between companies
Files sent directly to CAM machines and 3D printers	Risk of data loss, corruption, cyber-attack
Libraries of standard parts; quick communication worldwide	Slower than sketching for initial ideas; screen fatigue

Machine	How it works	Materials	Typical products
CNC lathe	Workpiece rotates in chuck; tools on a turret move in X/Z	Metals, polymers, wood	Shafts, bolts, knobs, pens
CNC router	High-speed spindle moves over a bed in X/Y/Z	Wood, MDF, plywood, foam, plastics, aluminium	Furniture parts, signs, moulds, sheet profiles
CNC milling machine	Rotating cutter; table/head move in 3–5 axes, automatic tool change	Metals, polymers	Engine parts, moulds, dies, precision components
CNC laser cutter	Focused CO_2 laser beam vaporises/burns material; also engraves	Acrylic, plywood, MDF, card, paper, fabric, leather (not PVC/ polycarbonate); fibre lasers cut metal	Signs, enclosures, packaging prototypes, jewellery
CNC vinyl cutter	Drag knife cuts vinyl film but not its backing	Self-adhesive vinyl, card, flock, heat-transfer vinyl	Signs, vehicle graphics, stickers, stencils, T-shirt transfers
Rapid prototyping (3D printing)	Additive: builds layer by layer	PLA, ABS, nylon, resins, metals (SLS/DMLS)	Prototypes, moulds, custom and complex parts

Key terms

CAD	Computer-aided design: using software to create, modify, analyse and present 2D drawings and 3D models (e.g. Fusion 360, SolidWorks, Onshape, 2D Design).
CAM	Computer-aided manufacture: using computers to control machines (CNC) that make products from CAD data.
CNC	Computer numerical control – machine movements controlled by coordinates (G-code) along X, Y and Z axes.
G-code	The numerical language of instructions (coordinates, feeds, speeds, tool changes) that CNC machines follow.
Virtual product	A photorealistic 3D CAD model that can be rendered, animated and tested before anything physical is made.
FEA	Finite element analysis – CAD simulation that applies loads to a virtual model and shows stress and weak points with a colour scale.
Rapid prototyping	Quickly making a physical model directly from CAD data, usually by additive manufacturing (3D printing) – FDM, SLA, SLS.
Nesting	Arranging parts close together on a sheet to minimise waste when laser cutting or routing.
Additive vs subtractive	Additive builds up material layer by layer (3D printing); subtractive removes material (milling, routing, turning, laser cutting).